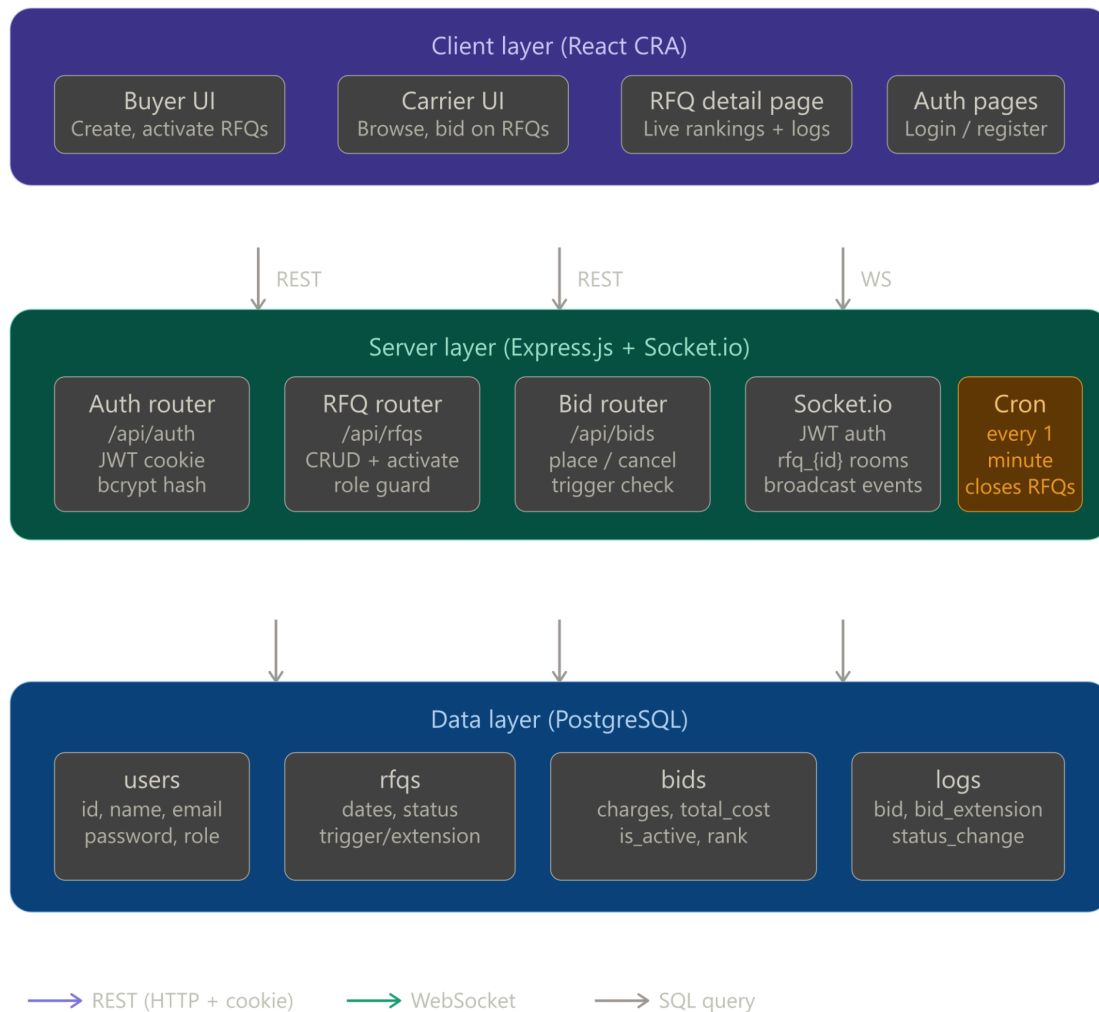


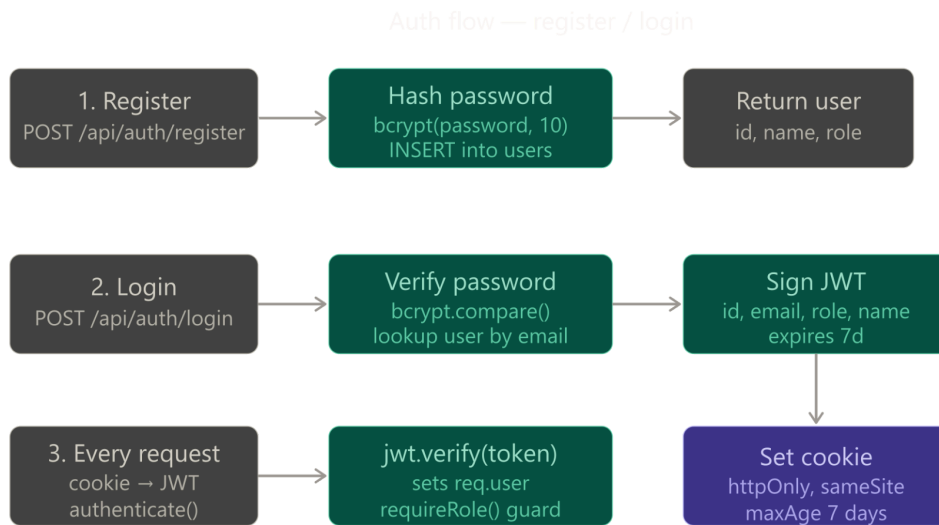
RFQ British Auction System

High Level Design — Architecture, Flows & Real-time Events

1. System Architecture Overview

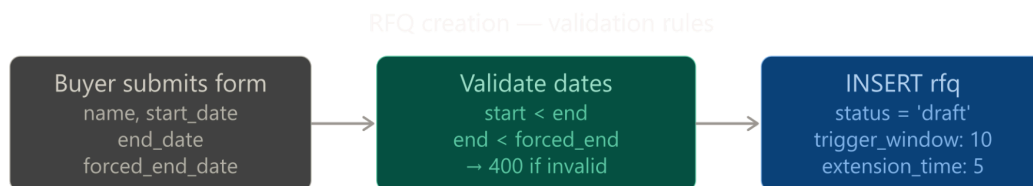
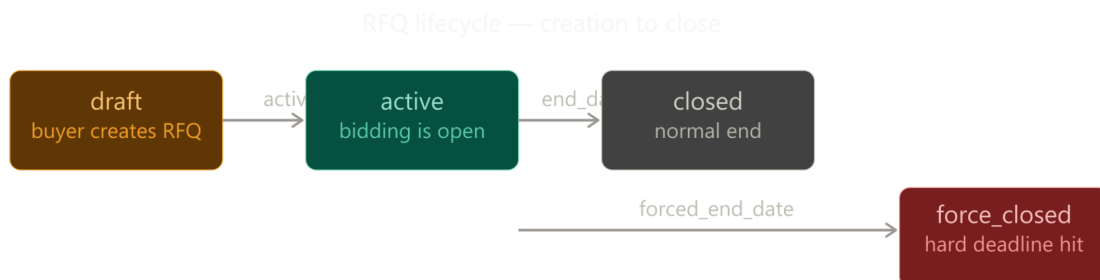


2. Authentication Flow

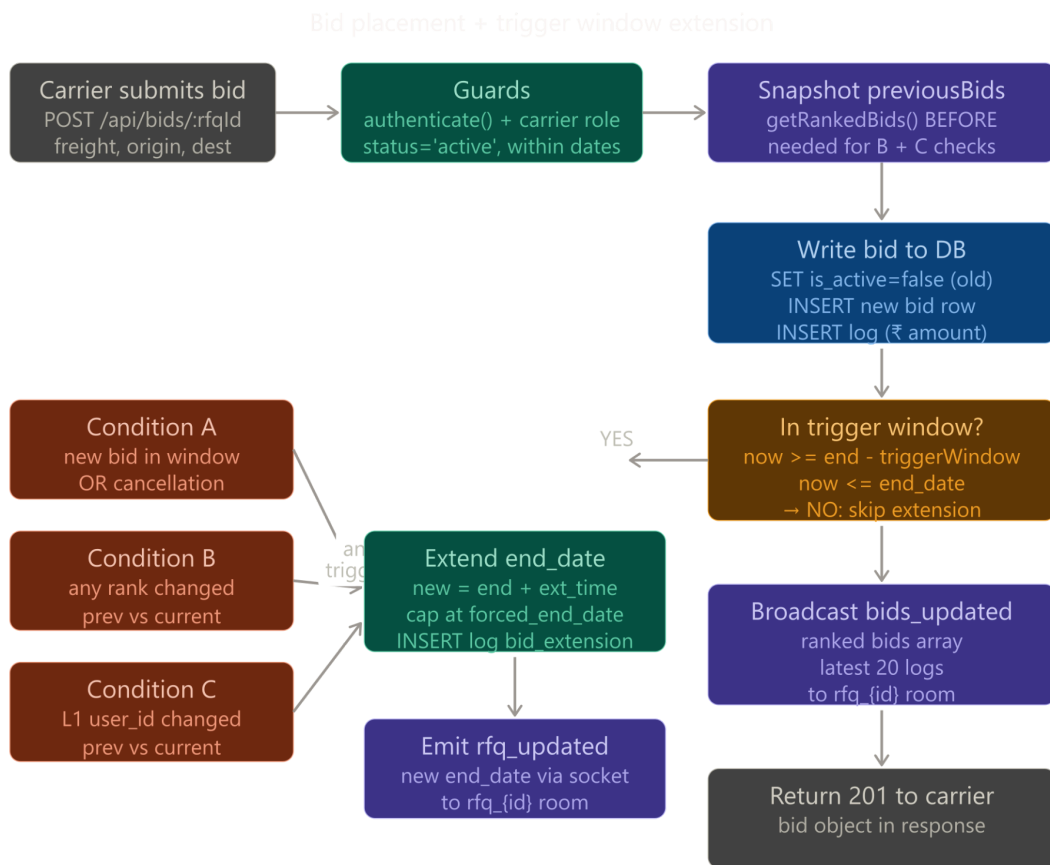


Cookie is httpOnly — JS cannot read it. Every API call uses credentials:'include'.
GET /api/auth/me restores session on page reload.

3. RFQ Lifecycle

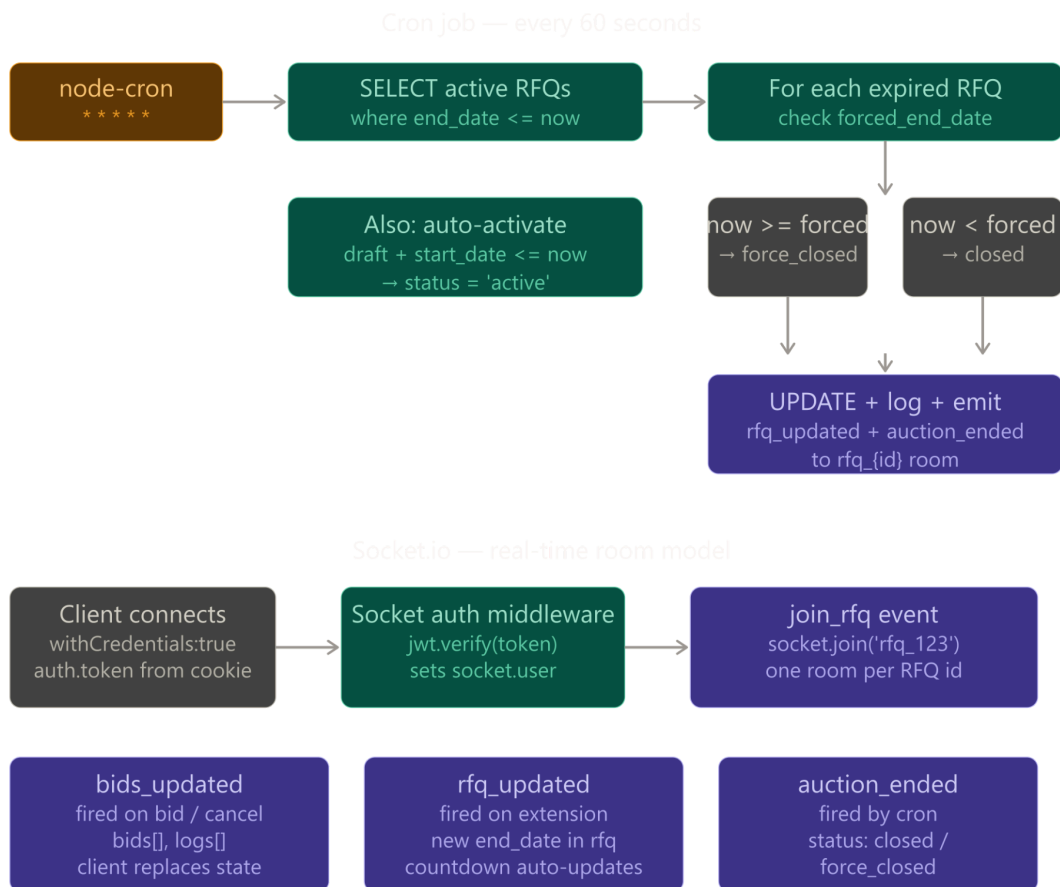


4. Bid Placement + Trigger Window Extension



Cancel bid (DELETE) follows identical path — snapshot → deactivate → condition check → broadcast.
Condition B compares every user_id rank before vs after. Condition C compares only bids[0].user_id.
Extension is capped: if new end > forced_end, forced_end is used instead. If already at cap, no extension.

5. Cron Job & Socket.io Real-time Events



Singleton socket on client — one connection shared across all pages. `leave_rfqs` called on page unmount.
`io.to('rfq_123').emit(...)` broadcasts only to clients viewing that specific RFQ — no unnecessary traffic.